DS4I Assignment 2: CNN part

Convolutional Neural Network

Methods

Question and task

- Question: can a writer be identified from a single handwritten digit?
- Input: one 28×28 image
- Output: a probability for each of the 20 writers (softmax); the predicted writer has the highest probability
- The digit is neither an input nor a label
- Same split as the Siamese network: train sessions 1–3, validation session 4, test session 5

Architecture

- Two convolutional layers: 16 filters of 3×3, each followed by an activation
- Dropout after the first layer, 2×2 max pooling, flatten
- 20-unit softmax output layer
- 48,580 trainable parameters, 95% of them in the output layer
- Loss: categorical cross-entropy; Adam optimiser; batches of 64

Models compared

- **Default CNN:** the course settings (ReLU, dropout 0.2, learning rate 0.01, no penalty)
- **Tuned CNN:** settings chosen on the validation session only
- **Nearest centroid:** each writer's average image of a digit as a template
- **Linear model:** softmax regression on the 784 pixels

Tuning: four stages, all on session 4

1. Activation (tanh or ReLU), learning rate, dropout rate: 72 combinations
2. Number of filters, kernel size, pool size: 12 combinations
3. Number of epochs, from the validation curves
4. L2 penalty ν: 8 values from 0 to 0.3

- Three random seeds per candidate
- Cross-entropy is computed without the penalty term, so values can be compared across penalties

Settings: default and tuned

Setting	Default	Tuned
Activation	ReLU	tanh
Dropout rate	0.2	0.7
Learning rate	0.01	0.0001
L2 penalty	none	0.1
Filters, kernel, pool	16, 3×3, 2×2	same
Epochs	10	10

How the models are judged

- Accuracy, cross-entropy, and per-writer recall and precision
- Every writer has 250 test images, so accuracy equals the average per-writer recall
- Only about 100 independent writer–digit cells per session (20 writers $\times$ 5 digits)
- So intervals come from resampling writers, not images

Results

Effect of the L2 penalty (validation session)

ν	Training accuracy	Validation accuracy	Validation cross-entropy
0	0.979	0.591	1.892
0.003	0.979	0.593	1.727
0.01	0.975	0.600	1.529
0.03	0.957	0.606	1.392
0.1	0.862	0.612	1.447
0.3	0.165	0.135	2.986

- U-shaped: the lowest cross-entropy is at 0.03; we used the next value, 0.1

Training versus validation

Model	Training	Validation	Gap
Default CNN	99.4%	44.8%	55 points
Tuned CNN	86.0%	60.4%	26 points

- The default CNN memorises the training sessions
- Tuning halves the gap, but a large gap remains
- The tuned score is optimistic: its settings were chosen on this session

Test performance (session 5)

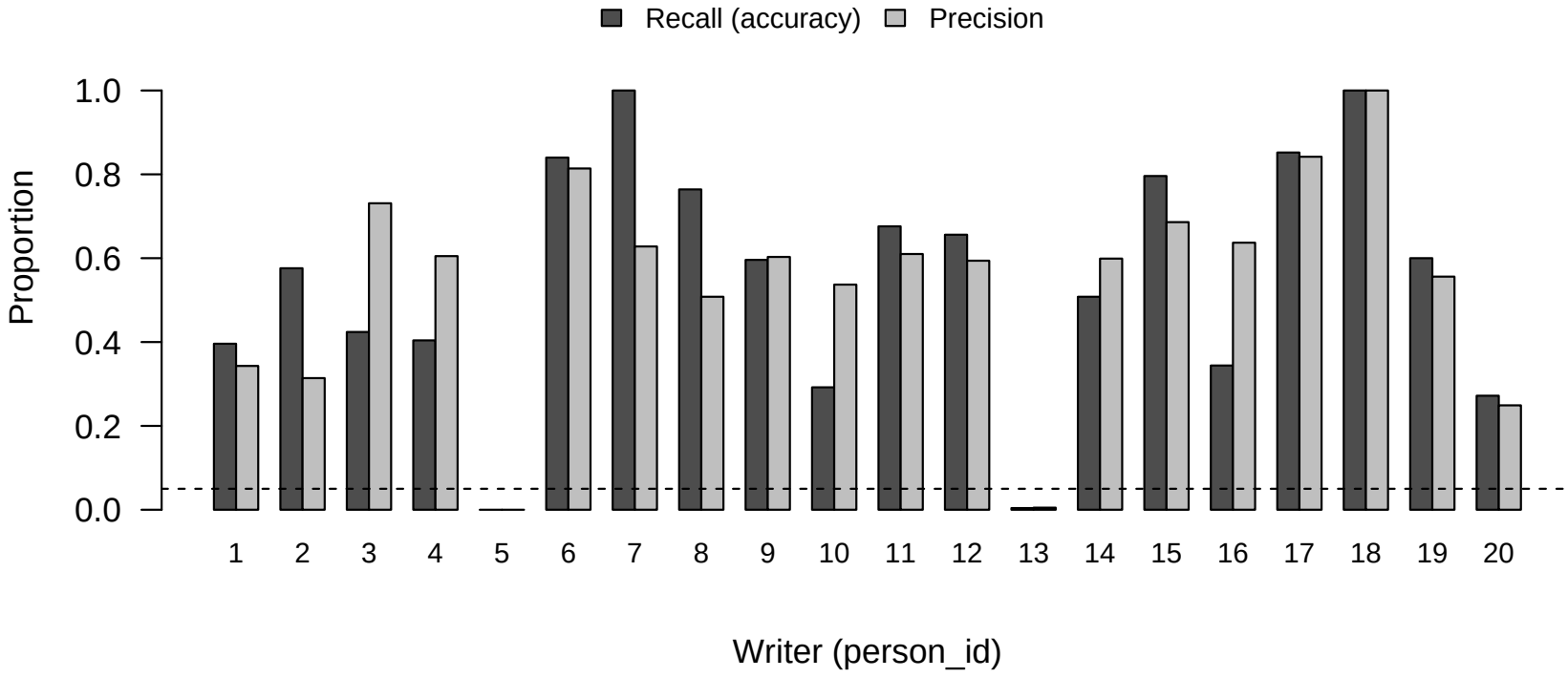
Model	Accuracy	Cross-entropy
Nearest centroid	47.3%	–
Linear softmax	43.6%	–
Default CNN	49.3%	5.87
Tuned CNN (final)	**55.0%**	**1.39**
Guessing among 20 writers	5.0%	3.00

- 95% interval for the tuned CNN: 42.7% to 67.1%

Is the tuned CNN really better?

- Versus the linear model: **+11.4 points** (95% interval 4.1 to 18.9), so unlikely to be due to which writers were sampled
- Versus the default CNN: $+5.7$ points (interval -1.9 to 13.6), which cannot be told apart from zero
- The clearest difference is in the probabilities: cross-entropy 1.39 against 5.87
- The default CNN is overconfident: wrong about half the time, but gives its wrong answers high probability

Accuracy by writer



- Writers 7 and 18: every image correct. Writer 5: none. Writer 13: 1 of 250

Where the errors go

- Writer 10 $\rightarrow$ writer 2: 71% of images (177 of 250)
- Writer 5 $\rightarrow$ writer 13: 60% (150 of 250)
- Writer 16 $\rightarrow$ writer 1: 40% (100 of 250)
- Writer 2 received 459 predictions, only 31% correct (a fair share is 250)
- Errors come in blocks of about 50: a whole writer–digit cell goes to one wrong writer
- Accuracy is similar for all five digits (50% to 59%)

Conclusions and limits

- A writer can be identified well above chance (55% against 5%), but not reliably
- A likely reason: a writer's average digit changes between sessions by about 80% of the distance to another writer's digit
- One validation and one test session, one training run per model, simulated data, a closed set of 20 writers
- The Siamese network answers a different question with different metrics, so the two are not directly comparable